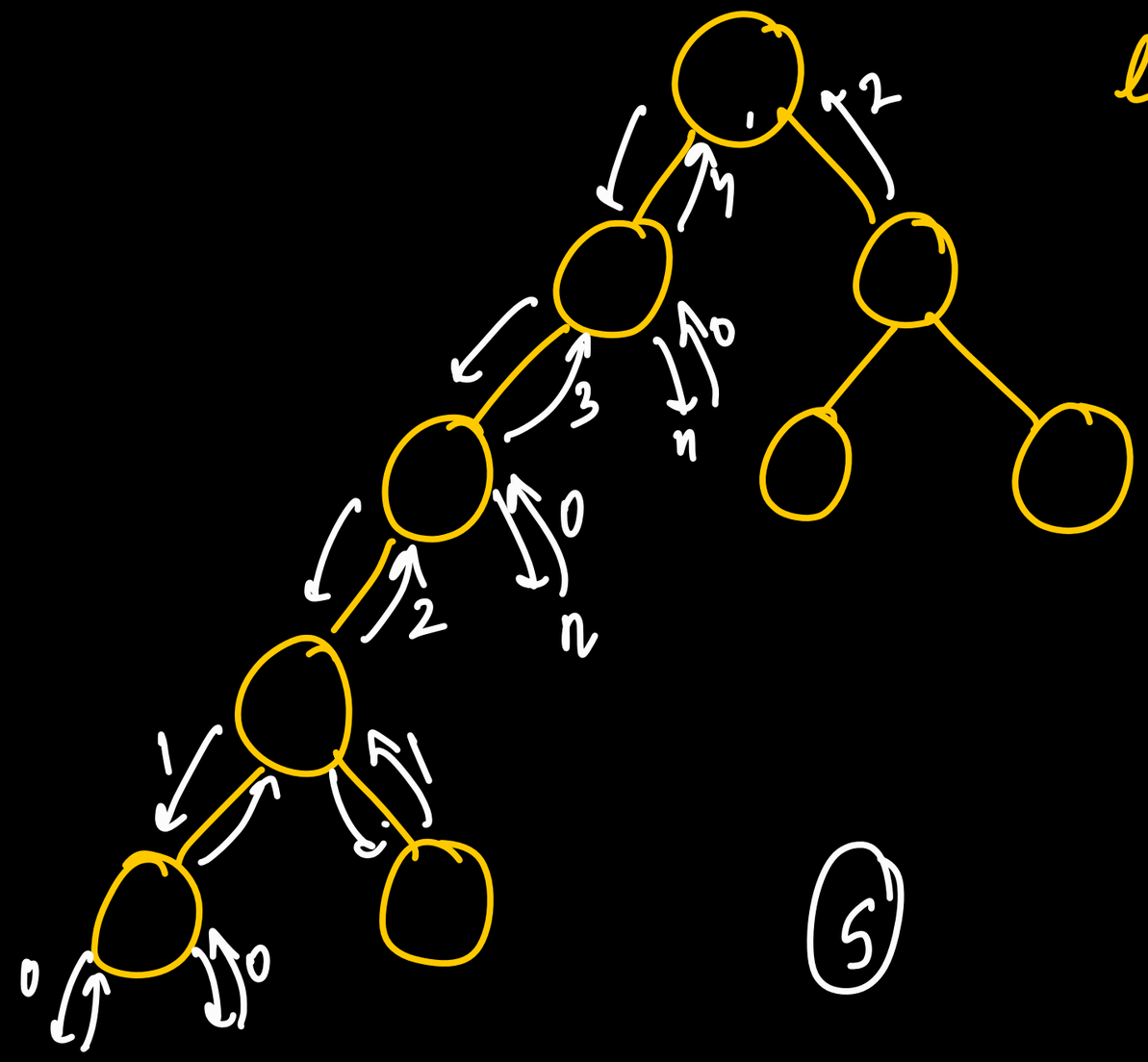
Ques: Levels of a Binary Tree

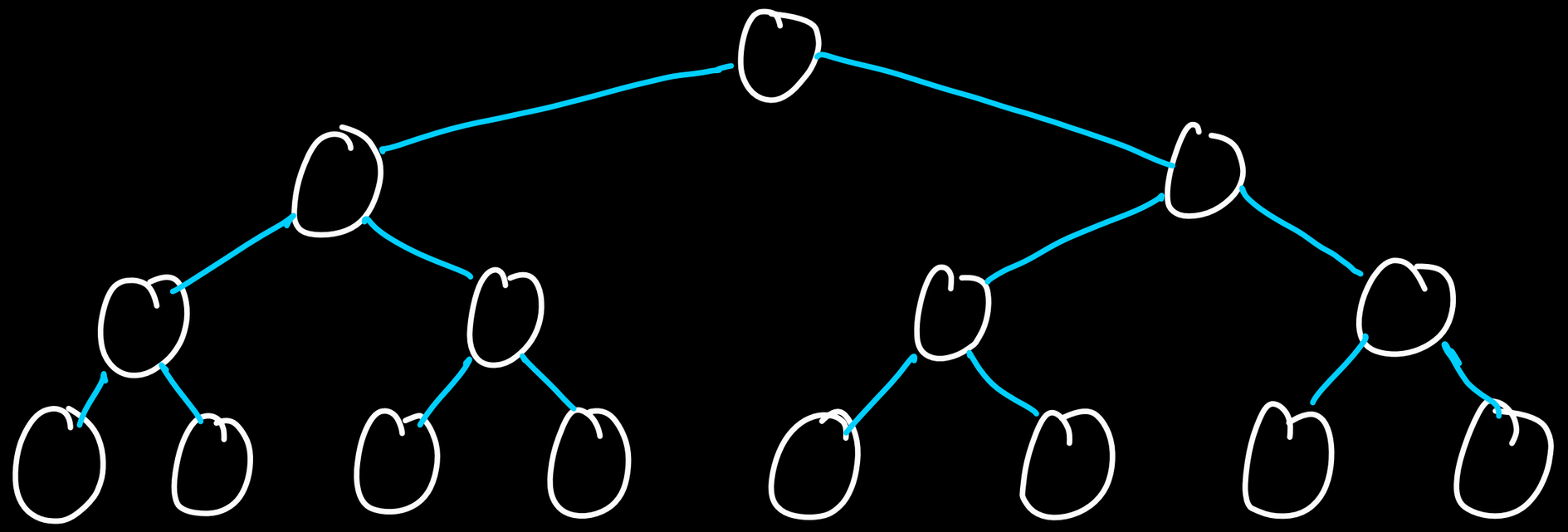


levels = 1 + max(levels(LST), levels(RST))

5

Ques: Levels of a Binary Tree

Size, sum, max



No. of calls $= \underbrace{1 + 2 + 4 + \cdots 2^{l-1}}_{\text{'}l\text{' levels}} = 2^{l} - 1$

$\boxed{l \approx \log_2 n}$

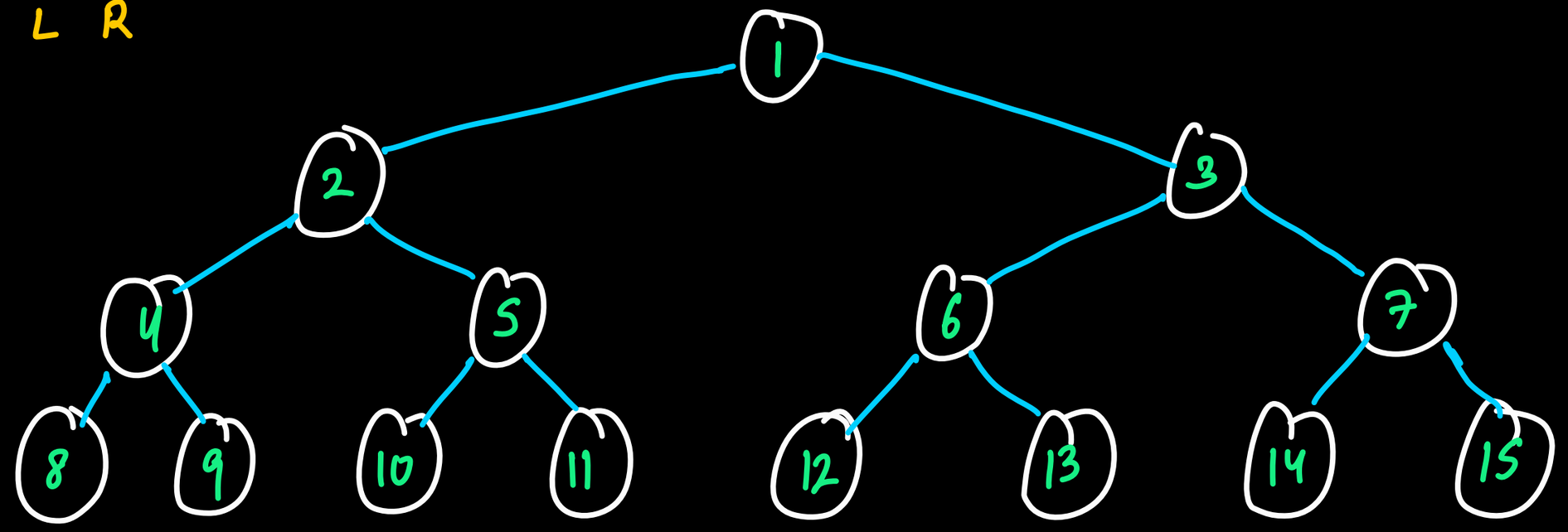
DFS Traversals

depth first search

Preorder	Node	Left	Right
Inorder	Left	Node	Right
Postorder	Left	Right	Node

Ques: Preorder traversal of a Binary Tree

Root L R



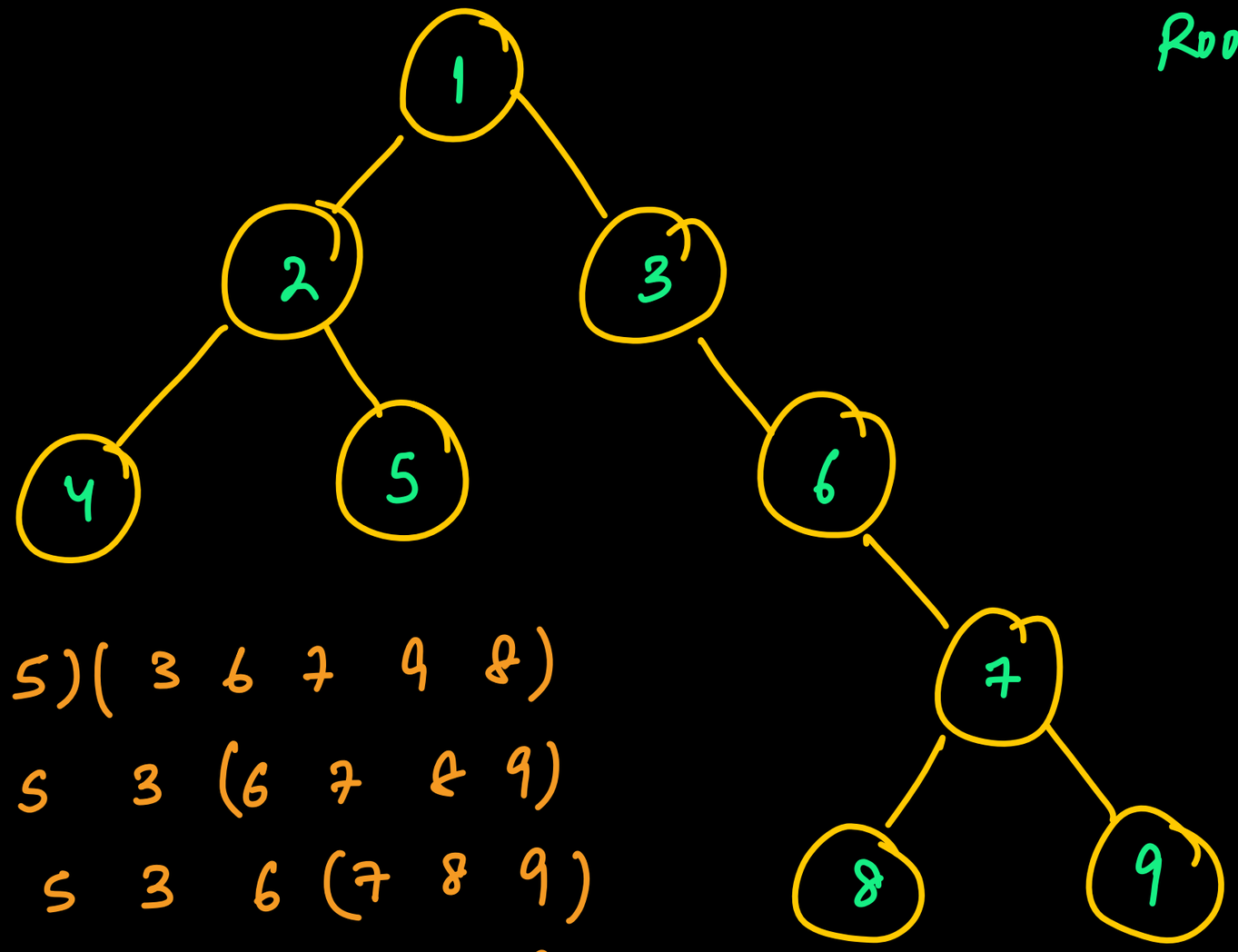
1 (2 4 5 8 9 10 11) (3 6 7 12 13 14 15)

1 2 (4 8 9) (5 11 10) 3 (12 13 6)(14 7 15)

1 2 4 8 9 5 10 11 3 6 12 13 7 14 15

Ques: Preorder traversal of a Binary Tree

Root left Right



1 (4 2 5)(3 6 7 9 8)

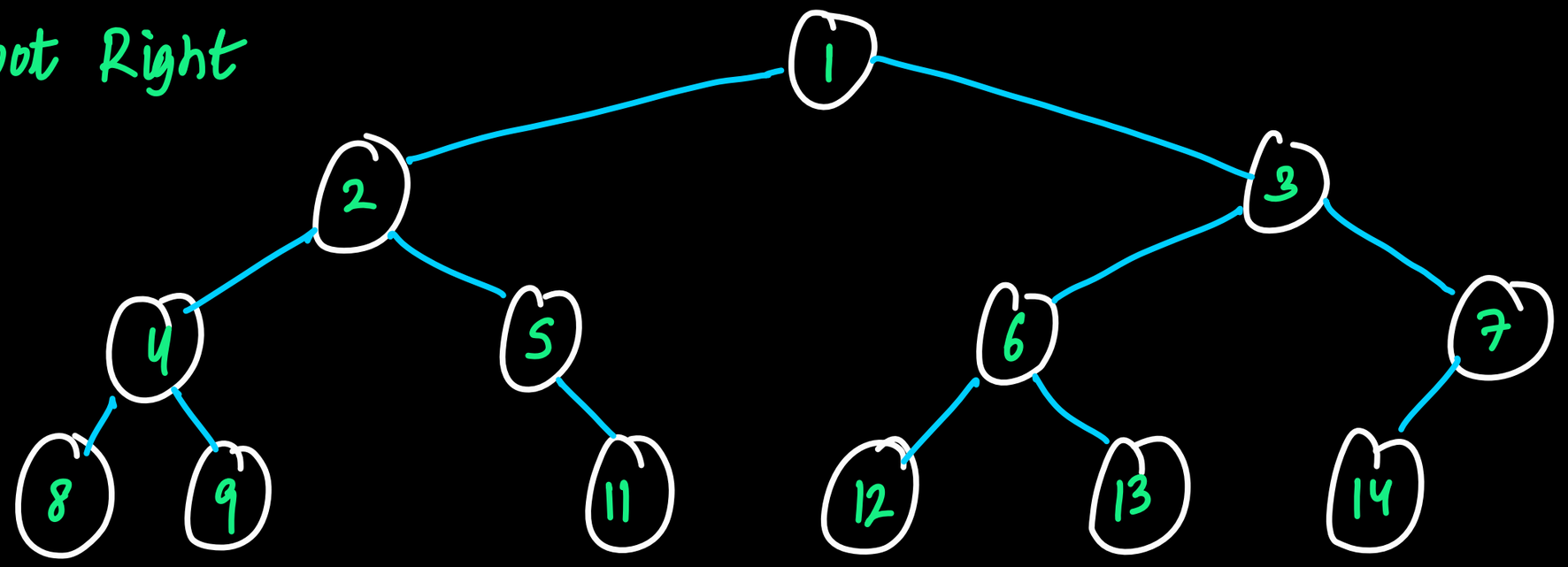
1 2 4 5 3 (6 7 8 9)

1 2 4 5 3 6 (7 8 9)

1 2 4 5 3 6 7 8 9

Ques: Inorder traversal of a Binary Tree

Left Root Right



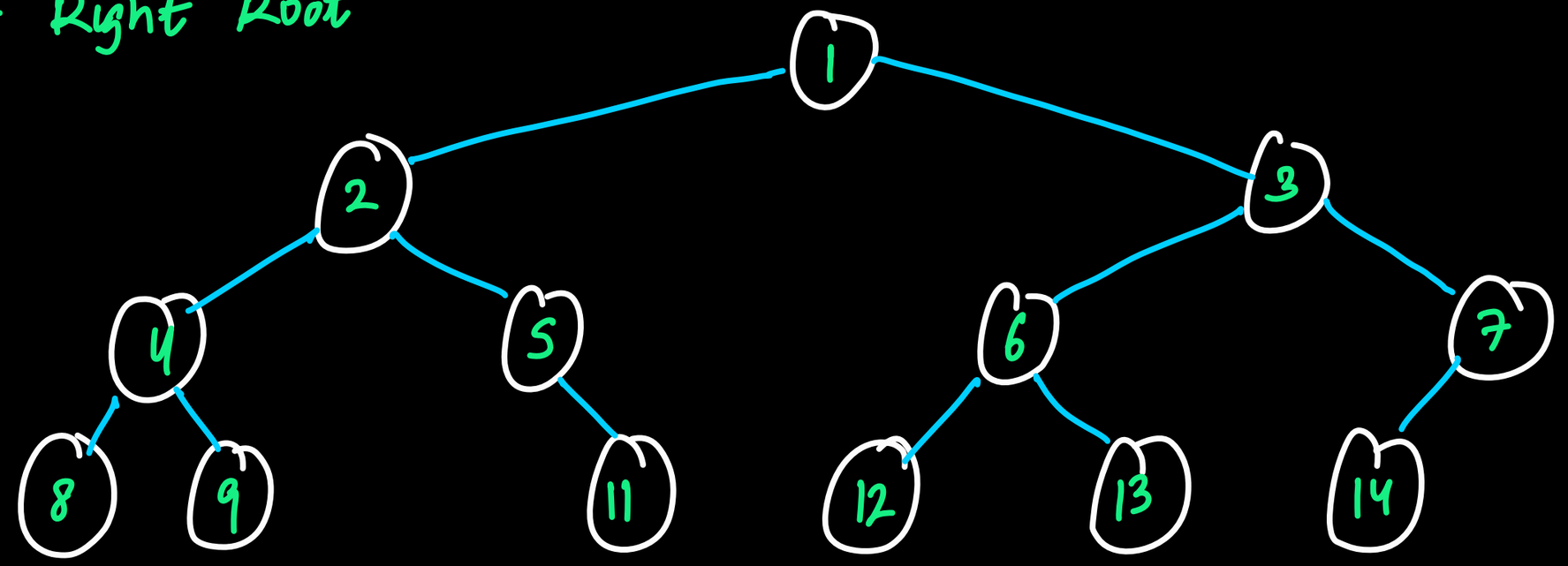
(2 4 8 9 5 11) 1 (3 6 12 13 7 14)

(4 8 9) 2 (5 11) 1 (6 12 13) 3 (7 14)

8 4 9 2 5 11 1 12 6 13 3 14 7

Ques: Postorder traversal of a Binary Tree

Left Right Root

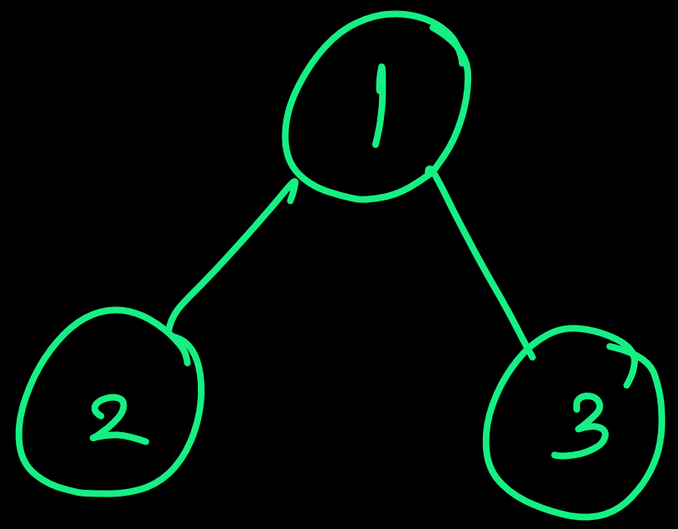


(2 4 5 8 9 11) (3 6 7 12 13 14) 1

(4 8 9) (5 11) 2 (6 12 13) (7 14) 3 1

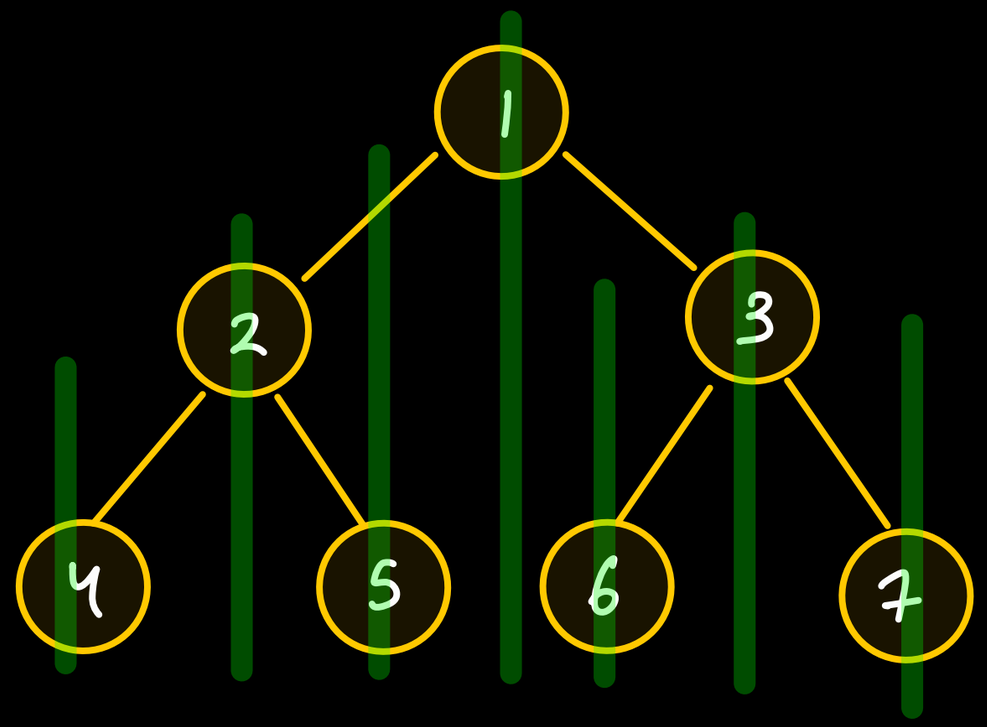
8 9 4 11 5 2 12 13 6 14 7 3 1

Baby Sawaal :



Pre	1	2	3
In	2	1	3
Post	2	3	1

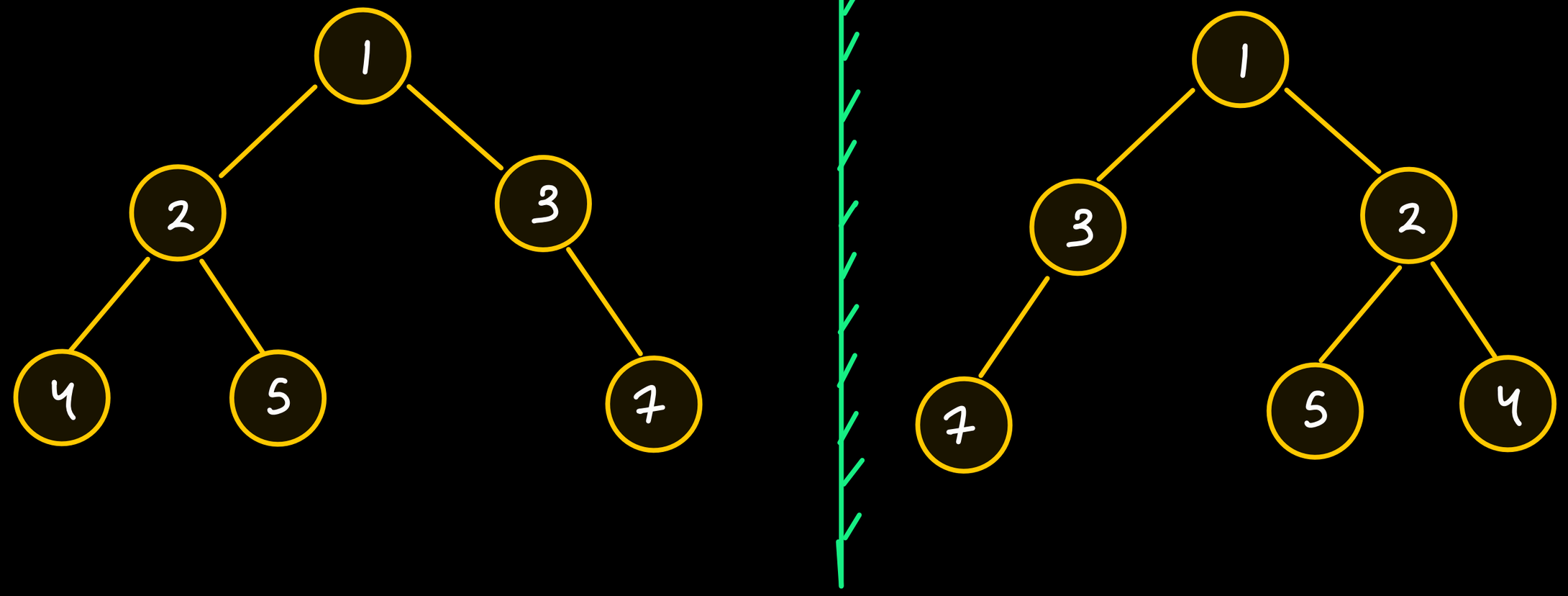
Euler's tour tree



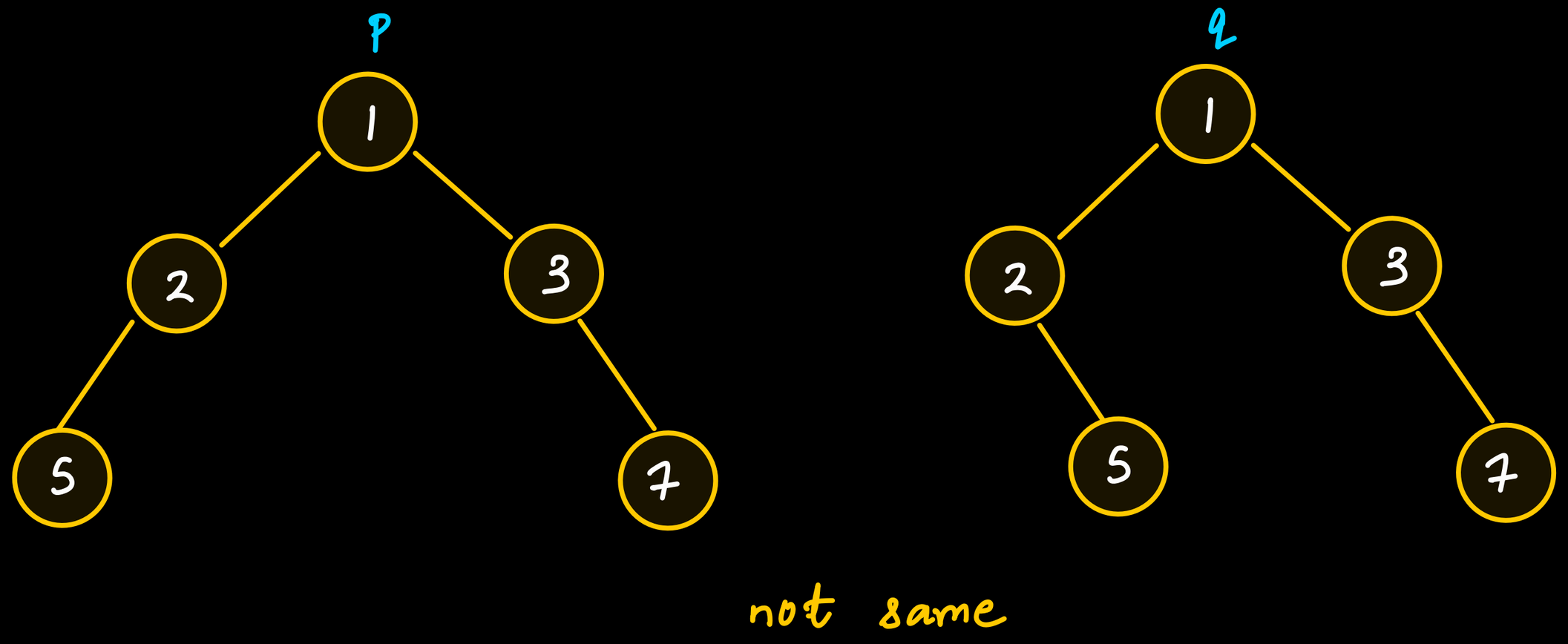
PostOrder : 4 5 2 6 7 3 1

Inorder : 4 2 5 1 6 3 7

Ques: Invert Binary Tree



Ques: Same Tree



not same

Ques: Symmetric Tree

Ques: Path Sum